US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

*Test Operations Procedure 2-2-540A 07 August 2024

TESTING OF UNMANNED GROUND VEHICLE (UGV) SYSTEMS

			Page
Paragraph	1.	SCOPE	2
	1.1	Purpose	2
	1.2	Applicability	2
	1.3	Activities Addressed	2
	1.4	Limitations	3
	1.5	Introduction to UGVs	3
	1.5.1	Need and Roles for UGVs	3
	1.5.2	UGV Categories	3
	2.	FACILITIES AND INSTRUMENTATION	5
	2.1	Test Facilities and Conditions	5
	2.2	Instrumentation	5
	2.3	Modeling and Simulation	6
	3.	REQUIRED TEST CONDITIONS	6
	3.1	Test Item Configuration	6
	3.2	Test Planning	7
	3.3	Frequency Allocation	8
	3.4	Supporting Documents	9
	3.5	Test Controls	11
	3.6	Safety	11
	4.	TEST PROCEDURES	13
	5.	DATA REQUIRED	23
	6.	PRESENTATION OF DATA	23
APPENDIX	A.	FREQUENCY ALLOCATION FORM	A-1
	B.	DOCUMENT REVIEW CHECKLIST	B-1
	C.	REFERENCES	C-1

*This TOP supersedes TOP 2-2-540, dated 12 February 2009.

Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

1. SCOPE.

This Test Operations Procedure (TOP) describes a systematic approach to safety and performance testing of Unmanned Ground Vehicles (UGVs). The objective is to ensure that the design of each UGV includes positive measures to enhance system safety, and that hazards which could reduce system safety are eliminated or controlled to an acceptable level of risk. This overarching document will point to other TOPs or International Test Operations Procedures (ITOPS) for more detailed explanation of specific test activities.

1.1 Purpose.

The purpose of this document is to provide a systematic approach to testing UGVs in order to ensure that the overall safety of the system is adequate and that the performance of the UGV meets expectations. It describes activities necessary to ensure safety is designed into the system under test. This document provides procedures for developing and implementing a comprehensive test methodology to identify hazards of a system and to verify performance of the system meets system requirements.

1.2 Applicability.

This document is appropriate for application to UGVs regardless of size, mode of operation, or type of payload employed. This document applies to tethered, remote-controlled, tele-operated, and autonomously controlled ground vehicles. Robotic systems addressed in this TOP cover the full spectrum, from throwable systems weighing approximately one quarter pound up to 70-ton vehicles, and range in capabilities from pure remote control or tele-operated systems to fully intelligent and autonomous systems. Any capabilities not under on-board human control such as weapons firing or vehicle movement are also covered by this document.

1.3 Activities Addressed.

The information contained herein applies to any testing of robotic systems, whether simply obtaining a safety release or conducting a formal developmental test program for systems acquisition. As with any test program, tailoring of test procedures to the specific needs of the system and its intended use are encouraged. This document also applies to the experimentation and demonstration of robotic system capabilities. Although many sections of this document may not apply for a particular system, safety related sections still apply, as the safety of personnel and avoidance of damage to vehicles, equipment, and facilities near robotic systems must be assured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

1.4 Limitations.

a. This document is only for testing of UGVs. Methodologies for testing of Unmanned Aerial Vehicles (UAVs) or Unmanned Submersible Vehicles (USVs) are not covered in this document. This document does not apply to automated subsystems that do not directly affect people or objects outside of the vehicle such as automated internal weapon loading systems or automated tracking subsystems, nor to automated driver-assist functions that require full-time on-board driver attention to perform a task such as adaptive cruise control, anti-lock brake systems, or self-leveling systems.

b. This document is applicable to testing of multiple UGVs working in the vicinity of each other; however, testing of multiple UGVs autonomously working in cooperation is not addressed in this document.

1.5 Introduction to UGVs.

1.5.1 Need and Roles for UGVs.

UGVs are robotic platforms that are used as an extension of human capability. These robots are generally capable of operating outdoors and over a wide variety of terrain, functioning in place of humans. UGVs are generally used to complete tasks that are, from a human perspective, dull, dirty, or dangerous. Tasks such as reconnaissance, explosive ordnance disposal, mine clearing, and logistic resupply are typical examples of jobs suited for UGVs. Due to the wide range of tasks that can be accomplished with UGVs, a wide range of types of UGVs are necessary; however, the general principles for testing UGVs can be applied regardless of UGV type.

1.5.2 UGV Categories.

UGVs can be generally defined based on three categories: size, mode of operation, and weapon type.

a. Size.

(1) MicroUGV. An unmanned ground vehicle weighing less than 10 lb.

(2) SUGV (Small Unmanned Ground Vehicle). An unmanned ground vehicle weighing less than 300 lb.

(3) MUGV (Medium Unmanned Ground Vehicle). An unmanned ground vehicle weighing between 300 and 2,000 lb.

(4) LUGV (Large Unmanned Ground Vehicle). An unmanned ground vehicle weighing more than 2,000 lb.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

b. Mode of operation.

(1) Manned. A mode of control wherein the UGV is directly controlled by a human operator through the use of a steering wheel, on-vehicle joystick, etc. This is considered a temporary mode of control for a system designed to be unmanned.

(2) Tethered. A mode of control wherein the human operator controls the UGV through a direct, wired connection. An example of such connection would be a fiber optic cable. Typically, a line of sight (LOS) is maintained under tethered operation; however, under certain circumstances, a LOS is not necessary (i.e., operation in tunnel, around corners, etc.).

(3) Remote Controlled. A mode of control wherein the human operator must dedicate their attention to system operation without benefit of sensory feedback from the vehicle. A LOS must be maintained with the vehicle under remote control operation.

(4) Teleoperated. A mode of control wherein the human operator has control of the UGV through cues provided by video, audio, and other digital feedbacks. The human operator controls the UGV through a wireless connection transmitted over radio frequencies (RF). The human operator must dedicate their attention to operating the UGV. A LOS does not necessarily need to be maintained under tele-operation.

(5) Partial Automation. A UGV that has multiple modes of control occurring simultaneously to include at least one autonomously controlled function. The level of semi-autonomy can vary greatly from UGV to UGV and will tend to be used extensively on weaponized UGVs (i.e., a weaponized UGV equipped with an Autonomous Navigation and Obstacle Avoidance System, but with tele-operated, operator-controlled weapon functions).

(6) Full Automation. A mode of control wherein the UGV is self-sufficient. The human operator can program a mission for the UGV, but the UGV can execute the mission without any human interaction.

c. Weapon type.

(1) Weaponized, projectile. A UGV equipped with any device that launches a projectile (i.e., machine gun, smoke grenades, unmanned ariel systems).

(2) Weaponized, non-projectile. A UGV equipped with any energetic device that can affect the area around the vehicle without launching a projectile (i.e., acoustic, laser, RF, sonic devices).

(3) Weaponized, non-lethal projectile. A UGV equipped with a weapon or device that launches a non-lethal projectile (i.e., rubber bullets, pepper balls, netting, paint balls, etc.).

(4) Non-weaponized. A UGV not equipped with a weapon, but may carry other types of payloads (smoke generator, sensors, targeting systems, bulk weight, etc.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

2. FACILITIES AND INSTRUMENTATION.

2.1 Test Facilities and Conditions.

a. Existing test facility types should be used when possible. It is necessary to consider all types of climatic and environmental conditions in a great variety of soil types. The tester should consider the effects of operating on various terrain including primary roads, secondary roads, cross-country terrain, and the ability to operate in and around natural and man-made obstacles. The tester should also consider the effects of high and low temperature extremes, humidity, rain, snow, ice, freezing rain, mud, sand, dust, vibration, and electromagnetic interference. Instrumentation and system operational equipment procedures for each facility should be created and may vary based on the type of equipment being tested.

b. Courses that are being used for UGV testing must be accurately digitally modeled in order to be used to verify the UGV decision making, when applicable. This digital model will be used as the "control" that can be compared to the UGV's perception in order to determine if there are variances between the UGV's perception and what actually exists. These facilities must be accurately instrumented and calibrated, to include RF and meteorological monitoring stations in order to accurately depict changing conditions of the course.

c. Dedicated UGV test courses should be equipped with complete video coverage of the entire course to record mishaps and incidents while also providing situational awareness to personnel monitoring the test from a safe location. For UGV operations in mixed-use areas, video coverage may be achieved with a camera mounted on the UGV.

d. RF spectrum should similarly be monitored and controlled, to prevent interference between multiple test items and infrastructure.

2.2 Instrumentation.

a. A wide range of instrumentation is available to measure the performance of the components, subsystems, sensors, algorithmic processes, and telemetric links, radio performance, and to capture required data. This data will be used not only to verify the performance and safety of the vehicle, but also to validate system models which can then be used to create larger data sets to complement physical testing. Therefore, when designing the instrumentation package for the system under test, data required to validate system models must be considered.

b. In general, instrumentation will continue to be used in traditional physical test roles to gather pertinent component level data using accelerometers, thermocouples, etc.

c. Additional instrumentation requirements must be considered when testing UGVs. UGVs with automation will rely extensively on sensors to provide information regarding the environment in which the UGV is operating. Data instrumentation packages should interface with the autonomy software to record the UGV's state, perceived obstacles, perceived location, and planned path.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

d. A redundant emergency stop (E-stop) with an independent power supply may be installed on each UGV if deemed necessary by test center personnel. Vehicle size, weight, speed, and existing safety devices should be considered in this decision.

e. Real Time and Near Real Time Analysis Displays. Testers need access to information to ensure the safety of all test personnel, to protect the test article as much as possible, and to ensure that the test unfolds as planned. Examples may include temperature profiles of critical components, comparisons of true and perceived vehicle location, vehicle planned paths, the states of weapons and payloads, actual and predicted data link quality, battery status, and other information as determined by the test director.

f. For Micro UGVs, SUGVs, and small MUGVs, the data acquisition/instrumentation package must be small enough so as to not affect the performance of the UGV under test.

2.3 Modeling, Simulation, and Analysis.

a. Data Link Prediction. Testers should use RF propagation prediction tools for some systems to predict the areas where the data link may be marginal or unacceptable for the test article and E-Stop systems. This information helps the tester maintain positive control of the UGV at all times and to ensure that a valid test is conducted.

b. 3D Models. As a part of test planning, testers may use 3D models of the vehicle and the test area to ensure that the vehicle is properly presented with challenges. Avenues of approach may be adjusted accordingly during the test planning process to achieve test objectives. These models may also be used during test execution to support monitoring UGV performance. This is especially important in areas of difficult terrain or where obscurants are present.

c. Modeling and simulation of components or entire UGV systems may be used to supplement, reduce, and/or focus real-world testing. The validity and accuracy of system software, behavior, and models are key to the accuracy of the simulation results. Modeling and simulation testing is outside of the scope of this TOP.

3. REQUIRED TEST CONDITIONS.

3.1 Test Item Configuration.

a. Sufficient testing should be conducted to demonstrate that the operator, software, vehicle platform, control unit, and mechanical operating devices and applications can work effectively together. Configuration control and recording of software versions is critical to successful conduct of developmental testing, as results of testing may only be valid for the exact configuration tested. In addition to system level testing, the subsystem and component level testing may be required. Any software update applied during test efforts should be accompanied with a justification of need, software configuration documentation, updated software and safety documentation and supporting evidence of the update’s intent and impact on the system. The test instrumentation suite should be selected based on the objective of the test and the data required. Individual components should be instrumented when required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

b. For a UGV, the main components addressed for system level testing are referred to in this document as the human operator, vehicle platform, software, Operator Control Unit (OCU), Mobile Base Unit (MBU) or Vehicle Control Unit (VCU), RF links, video links, tethered links, mechanical operating devices, electrical systems, E-stops, and payloads.

(1) The human operator controls the platform either remotely with an RF or tethered link or programs commands for autonomous operation.

(2) The vehicle platform provides mobility and electrical power.

(3) Software is used to control functions and coordinate the activities of other components.

(4) The OCU is the remote workstation for the human operator that enables remote control of selected driving, payload, and associated functions. This may also be referred to as the Human-Machine Interface (HMI), Warfighter Machine Interface (WMI), or similar terms.

(5) The MBU or VCU interface between the OCU and the mechanical operating devices on the UGV. RF, video, and tethered links establish communication between the UGV and the command vehicle.

(6) The application hardware is the means to perform the specified task of the UGV.

3.2 Test Planning.

a. Test planning should be initiated early in the acquisition process to yield the most cost-effective approach to decision risk mitigation. Component and subsystem level testing should be conducted throughout the development of the system. Each version must demonstrate effective support/operation of the system. System level testing should be conducted whenever an operational system is available. Performance of the individual components should be determined during system level testing to confirm design goals. All subsystems and components should be individually qualified by the system developer prior to integration into the system. Software is a critical component that must be rigorously tested and analyzed. ATEC Software Testers should participate in contractor-led software testing prior to system delivery to help focus the scope of testing.

b. Testing should be conducted to demonstrate that the logistics concept is adequate. Issues in the areas of maintenance, transportation, and storage must be addressed along with availability, accuracy, and comprehensibility. A logistics demonstration should be conducted with the system and representative Soldiers who will perform maintenance on the system. Transportability testing is required to obtain Transportability approval from the Military Surface Deployment and Distribution Command Transportation Engineering Agency (SDDTEA).

c. Test planning should address the following:

(1) Purpose of testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

(2) Test objectives.

(3) Test criteria.

(4) Test method.

(5) Instrumentation

(6) Data required.

(7) Data analysis techniques.

(8) Presentation of Data

d. Criteria for each subtest data topic will be defined based on applicable contractual specifications and military requirements documents. Where the criteria are not clearly specified, criteria will be developed from the relevant ITOP, TOP, military standard (MIL-STD), federal statute, or other controlling document. Development of criteria should consider the logistics concept, employment concept, intended application, novel technologies, and threat environment, as appropriate. Failures and successes should be defined to provide for a clear understanding of relevant conclusions for certain subtest data topics (i.e., reliability, performance, climatic suitability, electromagnetic interference, and safety).

e. Data requirements and analytical techniques necessary to address objectives and criteria will be established. It should be established which data requirements will be satisfied through physical test, and data sheets for each subtest data topic should be designed.

f. Appropriate test procedures for each subtest topic relative to objectives, criteria, and data requirements should be selected.

g. As a tool to audit, change, or summarize the test program, it is desirable to layout all events and data requirements in matrix format. As applicable, the test matrix should have a cell for each test objective, criteria, component, subsystem, system test event, item configuration, data requirement, and analytical technique. Multiple matrices, each of which is linked, may be required. Identify any unique resources necessary to conduct subtests.

h. Use of Modeling, Simulation, and Analysis to address test objectives should be identified within the detailed test plan.

3.3 Frequency Allocation.

a. The UGV Test Officer is responsible for requesting frequency assignments from the local frequency manager in the proposed area of deployment. The local frequency manager receiving the frequency assignment application will process it through established channels to request appropriate national and/or international approval. The disposition of the frequency assignment request will be channeled back to the user. The request may be approved, disapproved in total, or approved in part with operating limitations. The information presented in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

Appendix A of this document must be provided by the test proponent in order to submit the appropriate frequency allocation request form.

b. This information must be provided by the test proponent no less than 120 days prior to the start of testing to ensure approvals are obtained within the desired schedule. This information must be provided for each transmitter.

3.4 Supporting Documents.

a. To properly assess the safety, performance, and reliability of a UGV, a detailed documentation review must be conducted. This documentation review provides a more thorough understanding of UGV characteristics, developer's safety engineering efforts, completed verification, and in some instances can reduce the amount of physical testing required and can assist in reducing the assigned risk assessment code reported in safety releases. The documentation review is required to ensure that proper safety protocols are established prior to initiation of testing. The following documents should be included in the test item data package; it is preferable that these documents be provided as early as possible prior to the start of testing. A documentation review checklist is presented in Appendix B.

(1) Safety Assessment Report (SAR) - Required Document. The SAR is a formal, comprehensive safety report that summarizes the safety engineering efforts and safety data that have been collected and evaluated during the life cycle of an item. It expresses the considered judgment of the contractor or developing agency regarding the hazard potential of the item and any actions or precautions that are recommended to minimize these hazards and to reduce the exposure of personnel and equipment to them. In addition, Defense Centers for Public Health (DCPH) certifications must be provided for lasers, and Material Safety Data Sheets (MSDS) must be provided for batteries, fuels, etc. The delivered SAR will be assessed against MIL-STD-882E[1**].

(2) Preliminary Hazard Assessment (PHA) - Required Document. The PHA is a broad hazard-screening tool that includes a review of the work that will be performed in a facility and identifies the hazards associated with the work and the facility. The results of the PHA are used to determine the need for additional, more detailed hazard analysis; serve as a precursor documenting that further analysis is deemed necessary; and serve as a baseline hazard analysis where further analysis is not indicated.

(3) Software Requirements Specifications (SRS) – Required Document. The SRS specifies the requirements for a Computer Software Configuration Item (CSCI) and the methods to be used to ensure that each requirement has been met. The SRS contains functional as well as safety requirements.

(4) Capabilities Requirements Document (CRD)/Capability Development Document (CDD). The CRD/CDD describes the overall mission area, the type of system proposed, and the anticipated operational and support concepts in sufficient detail for program and logistics support planning and includes a brief summary of the mission need. If the mission need was not documented prior to preparation of the CRD/CDD, the process that investigated alternatives for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

satisfying the mission need and developing the operational requirements is explained.

(5) Performance Specification (PS)/System Specification (SS)/Purchase Description (PD). The PS/SS/PD communicates the user's requirements to the manufacturer. These documents translate operational requirements into more technical language that tells the manufacturer what will be considered an acceptable product and how it will be determined if the product is acceptable.

(6) Security Classification Guide (SCG) – Required Document. The SCG is a documentary form of classification guidance issued by an original classification authority that identifies the elements of information regarding a specific subject that must be classified and establishes the level and duration of classification for each such element.

b. Programmatic documents that were developed for the system, or those pre-existing documents that apply, should be used to identify test criteria. These documents define the mission scenarios, climatic conditions, operational, and electromagnetic environments in which the item must operate. Documents that should be used to establish test criteria include:

(l) Operational Mode Summary/Mission Profile.

(2) Capabilities Requirements Document/Capability Development Document.

(3) Purchase Description/Performance Specification/System Specification.

(4) Military Regulations, Standards, and other controlling directives.

(5) Federal Statutes.

(6) Test Directives.

c. The following documents are essential for the test personnel to understand the system under test and assess system safety according to MIL-STD-882E.

(l) Developer Software Functional Qualification Test Plans and Reports.

(2) Component Level Test Plans and Reports.

(3) System Operator Manuals.

(4) System Maintenance and Repair Manuals.

(5) Software Trouble/Problem Reports.

(6) System and Software Requirements

(7) Safety Significant Requirements List

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

(8) Software Traceability Analysis.

(9) Data Tap Locations and Interface Definitions.

(10) Software Design/Development Documents.

(11) Software Emulations.

(12) Message Protocols/Definitions.

(13) Programmatic Environmental, Safety, and Health Evaluation.

(14) Configuration Management Plan.

(15) Interface Control Document

**Superscript numbers correspond to those in Appendix C, References

3.5 Test Controls.

Test controls are listed for each subtest data topic in accordance with the applicable ITOP, TOP, or MIL-STD.

3.6 Safety.

l. Historically, UGVs have an elevated likelihood of uncontrolled movement caused by various means including RF interference, mechanical failures, software failures, complexity, etc. Additional precautions must be taken regarding tester safety that would not normally be required for a manned vehicle test where an operator has better situational awareness, immediate feedback, and direct access to vehicle controls. The functionality and behavior of individual components and subsystems should be demonstrated prior to conducting a system level test.

b. Additionally, a complete safety and health hazard analysis should be made throughout the course of the test. During the initial inspections and maintenance, observations should be made through visual examination and functional checks for compliance in design to assure personnel safety. During testing, any potential hazard observed during operations or maintenance actions must be recorded. All scheduled subtests should be considered in the overall safety and health assessment.

c. A complete and detailed analysis of system software, software development documents, software safety engineering, and software test results should be conducted prior to the start of actual physical testing.

(l) Early involvement by testers in system software development is required to validate the capabilities, robustness, and safety features of the software. A thorough understanding of the contractor's software development process, hazard

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

analysis, software test results, and software trouble reports will lead to an accurate assessment of the residual risk inherent in the system's software controls and processes.

(2) Clear, accurate documentation of software version updates/revisions and of regression testing of software version updates/revisions is required during all phases of testing.

d. Specific tests to determine the mechanical, software, and operating safety of the entire system can be completed only after all component subsystem testing has been completed or adequate provisions have been taken to prevent any accidental injury or damage. Components and subsystem safety testing should focus on verification of safety requirements.

e. The concept of a UGV locates the operator away from the platform (except in tethered mode); therefore, control must be maintained when the telemetric link between the remote operator and the vehicle is lost, an unanticipated event occurs, or an unsafe condition exists. At a minimum, the system must demonstrate the ability to safely stop when any uncontrolled event or unsafe condition occurs. All emergency and fail-safe systems must be tested, and their reliability must be determined.

f. In situations during testing when safety related decisions must be made, the wellbeing of personnel, buildings, equipment, and vehicles should be placed above that of the test item.

g. The number of personnel in the test area will be limited to those test critical personnel necessary to accomplish the test mission safely and efficiently. Observers and spectators will remain outside of the hazardous operating zone of the system at all times until the system is rendered safe and verified to be safe to approach.

h. Prior to shipping the test item, or transporting it between local test sites, a safety assessment must be performed. Specific items to be considered include loading and unloading procedures and transportability characteristics of the test item. UGVs operating in a robotic state will not be permitted access to test agency roadways, and therefore a plan must be in place to transition the test item from test site to test site.

l. Until the safety envelope has been determined by operating the item near the maximum safe limit, a thorough understanding of what the operator/maintainer has to do with, on, in, and around an item is unknown and critical hazards could exist. This is especially true of software-controlled systems, where unpredictable and unsafe responses may result from computer failure, maintenance interlocks, power failures, and power-up tests (DA PAM 73-1[2], paragraph 6-63).

j. The tester should identify the software components that control safety-related functions and give them special attention. Software safety activities should be initiated on that component and continued through the requirements, design, code analyses, and testing phases. The tester also might identify the need for a more formal evaluation of software safety, based on the probability that the software might cause or fail to prevent failures in a safety-critical system component (DA PAM 73-1, paragraph Q-26.b).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

k. Although operators may be proficiently trained before using a system, training is not a sufficient preventive measure or work-around in most cases for safety-related issues. Products of safety-related testing include documentation of conditions in which the system can and cannot be safely operated, and the propensity for the total system to enter unsafe operating conditions.

l. A safety release is required before any hands-on testing, training, use, or maintenance is performed by Soldiers. A safety release describes the specific hazards of the system, operational limits, required precautions, and residual risks that may remain (DA PAM 73-1, paragraph 6-64.b). The Army Evaluation Center authors the safety document, potentially with a recommendation from the test center containing test data. The safety release will have a specific start and end date and is only valid for a specific system configuration for a time period or event. A safety confirmation indicates if specific safety requirements are met, includes a risk assessment for hazards not adequately controlled, lists technical or operational limitations or precautions, and highlights safety problems that require further investigation (DA PAM 73-1, paragraph 6-65). The safety confirmation indicates the system is ready for fielding, has a specific start date, the residual risks and is valid indefinitely. The scope of testing to obtain a safety release and confirmation is dependent on a number of factors to include:

(1) Objectives of the test, demonstration, or experiment.

(2) Intended use of the robotic system.

(3) Types of payloads, attachments, or weapons associated with the system.

(4) Modes of control.

(5) Relative positions of people with respect to the system (spectators, casualties, remote operators, warfighters on the ground, manned vehicles, etc.).

(6) Functionality and reliability of the render safe system.

(7) Software safety engineering and software development processes.

(8) Analysis of witness testing, contractor software test results, and software trouble reports.

4. TEST PROCEDURES.

a. A list of test procedures and data required that should be considered when designing a comprehensive UGV test is presented in Table 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1. TEST PROCEDURES AND DATA
REQUIRED SOURCE DOCUMENTS

Subtest	Method	Description
Characteristics		
Preliminary Inspection	TOP 2-2-505A[3]	Provides inspection and baseline operation prior to initiation of testing. Ensures test item is safe and ready to initiate testing.
Physical Characteristics	TOP 2-2-500[4]	Determines dimensions and other physical characteristics of the test. Physical characteristics data are critical data used in mobility analyses and transportability.
Weight Distribution and Ground Pressure	TOP 2-2-801[5] ITOP 2-2-801(1)[6]	Determines weight distribution and ground pressure measurements. Ground pressure measurements are critical data used in mobility analyses, tire performance, and transportability.
Center of Gravity	TOP 2-2-800[7] ITOP 2-2-800(1)[8]	Provides information relative to roll stability, transportability, and input to mobility model programs.
Stowage	TOP 2-2-802[9]	Determines adequacy of on-board equipment stowage facilities in or on vehicles.
Transportability	ITOP 1-2-500(1)[10] TOP 1-2-500B[11]	Determines transportability characteristics of military equipment. Includes highway, marine, air (rotary and fixed wing), and rail transport as well as lift and tie-down provisions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Characteristics		
Rail Impact	TOP 1-2-500B	Determines structural integrity of the test item as well as the adequacy of the tie-down system and tie-down procedures.
Final Inspection	TOP 2-2-505A	Provides inspection and baseline operation following completion of testing
Performance		
Acceleration/Maximum Speed	TOP 2-2-602A[12] ITOP 2-2-602(1)[13]	Determines vehicle performance and safety characteristics while accelerating. Determines vehicle maximum safe speed capability.
Braking	TOP 2-2-608[14] ITOP 2-2-627(1)[15]	Determines vehicle performance and safety characteristics while braking.
Grades and Slopes	TOP 2-2-610A[16] ITOP 2-2-610(1)[17]	Determines vehicle performance and safety characteristics while operating on longitudinal and lateral slopes.
Steering and Handling	ITOP 2-2-609(1)[18] TOP 2-2-600[19]	Determines performance and safety characteristics of vehicle steering systems.
Standard Obstacles	TOP 2-2-611A[20] ITOP 2-2-611(1)[21]	Determines vehicle performance and safety characteristics while operating on standard obstacles to include gaps, walls, v-ditch, aircraft loading ramps, and frame twisting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Performance		
Fording	TOP 2-2-612A[22] ITOP 2-2-612(1)[23]	Determines vehicle performance and safety characteristics while fording. Includes shallow, deep, and submerged fording.
Line-of-Sight (LOS)/(Non-Line-of-Sight (NLOS)	TOP 2-2-543[24]	Determines operating and safety characteristics of tele-operated and remotely operated unmanned ground systems while operating at their maximum telemetric ranges.
Soft Soil Mobility	TOP 2-2-619A[25] ITOP 2-2-619(1)[26]	Determines vehicle performance and safety characteristics while operating in soft soils. Includes mud, sand, snow, swamps, wet clay, and rice paddies.
Winch and Tow Performance	TOP 2-2-712A[27]	Determines performance and safety characteristics of automotive winches.
Drawbar Pull	TOP 2-2-604[28] ITOP 2-2-604(1)[29]	Determines vehicle power available for acceleration, towing, or hill climbing.
Full-Load Cooling	TOP 2-2-607[30] ITOP 2-2-607[31]	Determines cooling characteristics of engine, power train, and auxiliary components of vehicles when subjected to full- and part-throttle operations, repeated steering maneuvers, and exposure to extreme environments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Software Performance	ITOP 1-1-056[32]	Used to evaluate a system's software functional capabilities. It does not specifically address other software-related issues, such as safety or security. The method for undertaking the software performance assessment discussed in this document addresses software T&E as an integral element of system T&E and is targeted at the system performance level. Key elements of this approach include the allocation of system requirements to software, assessment of software performance, and assessment of the impact of software on overall system performance.
Electric System Performance	TOP 2-2-601[33]	Provides procedures for evaluation vehicle electrical system performance including power supply for weapons and other subsystems.
Position Locations and Navigation System	TOP 6-2-598A[34]	Used to determine the performance and accuracy of integrated position location and navigation systems.
Automotive Driver Assist Systems	TOP 2-2-515[35]	Determines safety and performance characteristics of a vehicle's electronic driver assist technologies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Logistics		
Maintainability	TOP 2-2-503[36] ITOP 2-2-509(1)[37]	Determines the maintainability characteristics of vehicles.
Logistics Supportability	TOP 1-1-030[38]	Determines the logistic supportability of the test item through quantitative and qualitative analysis of the test item.
Reliability, Availability, Maintainability	ITOP 2-2-506[39] ITOP 2-2-509(1) TOP 2-2-506A[40]	Quantitative analysis of data collected during endurance testing in order to determine the overall ability of the system to complete tasks.
Fuel Consumption	TOP 2-2-603A[41] ITOP 2-2-603(1)[42]	Determines vehicle fuel consumption during both controlled and typical service operating environments
Safety/Software		
System Safety	TOP 1-1-060[43] TOP 2-2-508[44] JSSSA Implementation Guide for MIL-STD-882E[46]	Used to identify and evaluate hazards associated with test items. Testing will provide determination or assessment of personnel and equipment hazards in the system and associated operation and maintenance hazards.
Software Safety Testing and Analysis	ITOP 1-1-057[45] MIL-STD-882E JSSSA Implementation Guide for MIL-STD-882E	Describes the activities necessary to ensure that safety is designed into software that is acquired or developed and that safety is maintained throughout the software life cycle. It provides uniform procedures for developing and implementing a safety- critical software test methodology of sufficient comprehensiveness to identify the software caused hazards of a system and to impose design requirements and management controls to prevent mishaps. The objective is to ensure that the software design takes

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Software Safety Testing and Analysis (CONT)		positive measures to enhance system safety, and that software errors which could reduce system safety have been eliminated or controlled to an acceptable level of risk.
Teleoperated Unmanned Ground Vehicle Latency	TOP 2-2-546[47]	Used to identify the delay between image capture and display on an operator station.
Modeling, Simulation, and Analysis	ITOP 1-1-001[48] ITOP 1-1-002[49] TOP 2-2-549[50]	Used to verify accuracy and adequacy of system models and provides guidelines to ensure consistent documentation of the use of modeling and simulation technology in support of test and evaluation.
Fire Extinguisher System	TOP 10-3-001[51]	Determines performance characteristics of vehicle on-board fire extinguisher system.
Human Machine Interface		
Human Factors	TOP 1-2-610[52]	Used to provide human factors engineering assessment of equipment.
Noise Levels	TOP 01-2-608B[53]	Used to determine operating noise levels.
Toxic Fumes	TOP 2-2-614A[54]	Used to evaluate the level of toxic fumes present at an operator location.
Environmental		
Environmental Performance - High Temperature - Low Temperature - Temperature Shock - Solar Radiation - Rain and Snow - Humidity - Fungus - Salt Fog - Sand and Dust - Icing/Freezing Rain - Shock Vibration	TOP 2-4-001[55] TOP 2-4-002A[56] MIL-STD-810H[57]	Determines the operating, maintenance, and durability characteristics of unmanned ground vehicles when operating in extreme environments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Electromagnetic Interference (EMI)/Electromagnetic Compatibility (EMC) - Radiated Emissions - Radiated Susceptibility - Hazards of Electromagnetic Radiation to Ordnance (HERO) - Hazards of Electromagnetic Radiation to Fuel (HERF) - Hazards of Electromagnetic Radiation to Personnel (HERP) - Helicopter Electrostatic Discharge (HESD) - Personnel Electrostatic Discharge (PESD) - High-Altitude Electromagnetic Pulse (HEMP) - Near Strike Lightning (NSL) - Bonds and Grounds - Conducted Susceptibility - Conducted Emissions - Intra-system EMC	TOP 1-2-511A[58] TOP 2-2-613[59]	Determines whether the item tested meets the electromagnetic radiation effects, static electricity, and lightning criteria and the maximum electromagnetic radiation environment to which the test item may be exposed without adverse effects. Ensures that the equipment under test can operate in its intended electromagnetic environment without its performance being degraded and without degrading the performance of other system(s) in close proximity
Survivability		
Nuclear Survivability	TOP 1-2-612A[60]	Determines the effects of a specified nuclear environment on an unmanned ground vehicle.
Armor/Survivability	ITOP 2-2-617[61] ITOP 4-2-508[62]	Determines the vulnerability of unmanned ground vehicles and their components and subsystems to various levels of threats including, but not limited to IEDs, mines, etc.
Signature Measurements	TOP 02-2-812B[63]	Determine the infrared signature of a ground target.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Chemical and Biological Contamination Survivability	TOP 08-2-111C[64]	Used to determine the survivability of small items of equipment after exposure to chemical or biological hazards.
Weapons		
Boresighting	ITOP 3-2-836(2.1.1)[65]	Detects angular changes between gun and sighting systems, with or without the aid of an MRS, after vehicle operations over cross-country courses and primary and secondary roads, after firing, and after a period of temperature changes.
Computer Solutions	ITOP 3-2-836(2.4.1)[66]	Non-firing tests that determine the differential angle between the main weapon and sight line (ballistic solution) generated by a computerized fire control system in response to given inputs.
Fields of Fire	TOP 3-2-813[67]	Determining the given area which a weapon or group of weapons can cover effectively with fire from a given position for vehicle-mounted primary and secondary armament. This includes maximum elevation and depression angles at all traverse positions; maximum traverse angles or provisions for continuous traverse; and minimum range of anti-personnel fire at all positions of traverse.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
Frequency Response	ITOP 3-2-836(1.3.2.3)[68]	Determines the frequency response characteristics of UGV gun/turret drive systems in the turret whether they are installed on UGVs or on test stands.
Rangefinder	ITOP 5-2-618[69]	Evaluates the performance and safety characteristics of laser rangefinders under a variety of expected operating conditions.
Weapon Firing/Accuracy	ITOP 3-2-836(2.2.1)[70]	Determines the unintended deviation of the LOS of a stabilized gun/turret and a stabilized sighting system.
Target Acquisition and Tracking	ITOP 3-2-836(2.3.5)[71]	Used to measure the target tracking capability of a UGV mounted optical or electro-optical sighting system operated by an automatic system, under specified conditions in the laboratory and in the field.
Sight Plumb & Synch	ITOP 3-2-836(2.1.2)[72]	Evaluates the capability of vehicle-mounted main-gun sighting systems.
Stabilization	ITOP 3-2-836(1.3.2.2)[73]	Determines the stabilization performance of the UGV gun/turret drive systems in the turret whether they are installed on the UGVs or on the test simulators.
Time-of-fire, Rate-of-fire	ITOP 3-2-836(2.3.4)[74]	Used to determine times necessary to complete various aspects of the target engagement sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

TABLE 1 (CONT)

Subtest	Method	Description
EMI		
Interference	MIL-STD-461E[75]	Provides verification requirements for the control of the EMI (emission and susceptibility) characteristics of electronic, electrical, and electromechanical equipment and subsystems.
Interface requirements	MIL-STD-464D[76]	Establishes electromagnetic environment effects interface requirements and verification criteria for UGVs.
Additional UGV TOPs	TOP 2-2-541[77] TOP 2-2-542[78]	Additional TOPs covering Safe Operation of Mobile UGVs and Weaponized UGVs.

5. DATA REQUIRED.

The data to be acquired for the testing of UGVs is outlined in Section 5 of each TOP in Table 1.

6. PRESENTATION OF DATA

a. Describe the inspection, specific test procedures, and results for each item using narration, tables, photographs, x-rays, charts, and graphs as appropriate or as outlined in the procedures specific methodology.

b. Reduce, summarize, and analyze data from each subtest appropriate to the subtest data topic and failure definitions derived specifically for the item and the subtest category. When unique analytical tools (i.e., models, simulations, statistical techniques) are used, these should be described in sufficient detail to enable the reader to understand the basis for the analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX A. FREQUENCY ALLOCATION FORM

(1) Name of Program
(2) Purpose of Program
(3) Security Classification
(4) Frequency
(5) Transmitter Power (Watts)
(6) Time of Usage
(7) Required Start and End Dates
(8) Transmitter Nomenclature
(9) Transmitter Location
(10) Transmitter Antenna Data (a) Type/Name (b) Gain (c) Site Elevation (d) Antenna Feedpoint Height (e) Orientation (f) Polarization
(11) Receiver Nomenclature
(12) Receiver Location
(13) Receiver Antenna Data (a) Type/Name (b) Gain (c) Site Elevation (d) Antenna Feedpoint Height (e) Orientation (f) Polarization
(14) J/F-12 Number (if assigned)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX B. DOCUMENT REVIEW CHECKLIST

ATEC Test Checklist (Reference DA PAM 73-1)

a. In order to properly assess the safety, performance, and reliability of a UGV, a detailed documentation review must be conducted. This documentation review provides a more thorough understanding of UGV characteristics and in some instances can reduce the amount of physical testing required and can assist in reducing the assigned risk assessment code reported in safety releases. The documentation review is required to ensure that proper safety protocols are emplaced prior to initiation of testing. The following documents should be included in the test item data package; it is preferable that these documents be provided as early as possible prior to the start of testing.

1. Safety Assessment Report (SAR).

2. Preliminary Hazard Assessment (PHA).

3. Software Requirements Specification (SRS).

4. Capabilities Requirements Document (CRD)/Operational Requirements Document (ORD).

5. Performance Specification (PS)/System Specification (SS)/Purchase Description (PD).

6. Security Classification Guide (SCG).

7. Operational Mode Summary (OMS)/Mission Profile (MP).

8. Military Regulations, Standards, and other controlling directives.

9. Federal Statutes.

10. Test Directives.

11. Developer Software Functional Qualification Test Plans and Reports.

12. Component Level Test Plans and Reports.

13. System Operator Manuals.

14. System Maintenance and Repair Manuals.

15. Software Trouble/Problem Reports.

16. System Safety – Critical Software Requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

17. Software Traceability Analysis.

18. Data Tap Locations and Interface Definitions.

19. Software Design/Development Documents.

20. Software Emulations

21. Message Protocols/Definitions.

22. Programmatic Environmental, Safety, and Health Evaluation.

23. Configuration Management Plan.

24. Interface Control Document (ICD).

25. Health Hazard Assessment Report (HHAR).

26. Safety Data Sheet

27. All test data available regarding the item requiring the Safety Release. If no current test data are available, any other information that can be used (for example, prior Government test data, contractor test data), with the emphasis on safety data.

28. Training plans.

29. Equipment publications.

30. Test and Evaluation Master Plan (TEMP).

31. Frequency Allocation Documentation.

32. USAPHC Certifications (Laser and High Power RF).

33. Source of troops involved in operational testing.

34. Test Readiness Review (TRR).

b. When sufficient data are not available on which to base a Safety Release, additional testing may be necessary. In such cases, required testing will be performed by ATEC, and test costs will be paid by the materiel developer; the time required for issuing a Safety Release would increase accordingly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX C. REFERENCES.

1. MIL-STD-882E, Department of Defense Standard Practice for System Safety, 11 May 2012.

2. Department of the Army Pamphlet (DA PAM) 73-1, Test and Evaluation in Support of Systems Acquisition, 30 May 2003.

3. Test Operations Procedure (TOP) 2-2-505A, Inspection and Preliminary Operation of Vehicles. 22 October 2018.

4. TOP 2-2-500, Vehicle Characteristics, 14 February 2008.

5. TOP 2-2-801, Weight Distribution and Ground Pressure (Wheeled and Tracked Vehicles), 26 September 2006.

6. International Test Operations Procedure (ITOP) 2-2-801(1), Tracked Vehicle - Weight Distribution & Ground Pressure, 15 May 1987.

7. TOP 2-2-800, Center of Gravity, 26 September 2006.

8. ITOP 2-2-800, Tracked Vehicle Center of Gravity 15 May 1987.

9. TOP 2-2-802, Stowage, 9 January 1979.

10. ITOP 1-2-500, Tracked Vehicle Transportability, 18 May 1987.

11. TOP 1-2-500B, Transportability, 23 February 2022.

12. TOP 2-2-602A, Acceleration; Maximum and Minimum Speeds, 25 March 2019.

13. ITOP 2-2-602(1), Tracked Vehicle Acceleration, Maximum and Minimum Speeds, 21 May 1987.

14. TOP 2-2-608, Braking, Wheeled Vehicles, 20 May 2008.

15. ITOP 2-2-627(1), Tracked Vehicle – Braking, 21 May 1987.

16. TOP 2-2-610A, Gradeability and Side – Slope Performance, 20 April 2020.

17. ITOP 2-2-610(1), Tracked Vehicle – Gradeability & Side-Slope Performance, 21 May 1987.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

18. ITOP 2-2-609(1), Tracked Vehicle – Steering, 18 May 1987

19. TOP 2-2-600, Steering Performance, Tactical Vehicles, 29 July 2015.

20. TOP 2-2-611A, Standard Obstacles, 13 May 2022.

21. ITOP 2-2-611(1), Standard Obstacles, 21 May 1987.

22. TOP 2-2-612A, Fording, 8 May 2019.

23. ITOP 2-2-612(1), Tracked Vehicle – Fording, 18 May 1987

24. TOP 2-2-543, Line-Of-Sight/Non-Line-Of-Sight Testing of Unmanned Ground Vehicle (UGV) Systems, 24 November 2008

25. TOP 2-2-619A, Soft-Soil Vehicle Mobility, 11 February 2016.

26. ITOP 2-2-619(1), Tracked Vehicle - Soft Soil Mobility, 01 June 1987.

27. TOP 2-2-712A, Automotive Winches, 6 April 2022.

28. TOP 2-2-604 CN1, Drawbar Pull, 26 January 2017.

29. ITOP 2-2-604(1), Tracked Vehicle – Drawbar Pull on Soft Soil, 20 May 1987, w/ch1, 11 August 1987.

30. TOP 2-2-607, Cooling Systems (Automotive), 27 September 2006.

31. ITOP 2-2-607(1), Tracked Vehicle – Full Load Cooling, 21 May 1987.

32. ITOP 1-1-056, Software Performance Testing and Analysis, 27 July 2010.

33. TOP 2-2-601, Electrical Systems (Vehicles and Weapon Subsystems), 20 June 1977.

34. TOP 6-2-598A, Position Location and Navigation Systems (PLANS), 18 June 2013.

35. TOP 2-2-515, Advanced Driver Assistance Systems, 22 August 2022.

36. TOP 2-2-503, Maintenance (Vehicle), 5 June 1966.

37. ITOP 2-2-509(1), Tracked Vehicle Reliability, Availability and Maintainability, 28 May 1987, w/ch1 11 August 1987.

38. TOP 1-1-030, RAM-D and ILS Analysis, 8 September 2008.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

39. ITOP 2-2-506(1), Tracked Vehicle Endurance Testing, 15 May 1987.

40. TOP 2-2-506A, Endurance Testing of Tracked and Wheeled Vehicles, 2 October 2014.

41. TOP 2-2-603A, Vehicle Fuel Consumption, 10 May 2012.

42. ITOP 2-2-603(1), Tracked Vehicle Fuel Consumption, 18 May 1987.

43. TOP 1-1-060, System Safety Engineering, 12 September 1986.

44. TOP 2-2-508, Automotive Safety and Health Hazard Evaluation, 24 November 1982.

45. ITOP 1-1-057, Safety Critical Software Analysis and Testing, 20 July 2010.

46. Joint Software System Safety Engineering Handbook (JSSSEH), Version 1, 27 August 2010.

47. TOP 2-2-546, Teleoperated Unmanned Ground Vehicle (UGV) Latency Measurements, 11 January 2017.

48. ITOP 1-1-001, General Guidance for Incorporating Modeling and Simulation in Test Operations Procedures, 6 September 2004.

49. ITOP 1-1-002, General Procedure for Modeling and Simulation Verification and Validation information Exchange, 13 May 2004.

50. TOP 2-2-549, Modeling and Simulation (M&S) of Autonomous Unmanned Ground Vehicle (UGV) Systems, 14 June 2023.

51. TOP 10-3-001, Performance and Vulnerability Testing of Crew/Interior Compartment AFES Used In Combat and Tactical Vehicles, 6 February 2012.

52. TOP 1-2-610, Human Factors Engineering (Test Procedure), 15 May 1990.

53. TOP 1-2-608B, Sound Level Measurements, 4 March 2019.

54. TOP 2-2-614A, Toxic Hazards Test for Vehicles and other Equipment, 14 May 2020.

55. TOP 2-4-001, Desert Environmental Testing of Wheeled and Tracked Vehicles, 12 May 1969.

56. TOP 2-4-002A, Arctic Environmental Test of Tracked and Wheeled Vehicles, 11 December 2015.

57. MIL-STD-810H, Environmental Engineering Considerations and Laboratory Tests, 31 January 2019.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

58. TOP 1-2-511A, Electromagnetic Compatibility Requirements, Systems Testing, 20 November 2013.

59. TOP 2-2-613, Electromagnetic Interference Testing for Vehicles and Electrical Subsystems – Non-Communications, 16 December 1994.

60. TOP 1-2-612A, Nuclear Environment Survivability, 28 October 2014.

61. ITOP 2-2-617, Vulnerability Testing of Combat Vehicles and Their Components/Subsystems (Utilizing Conventional Weapons), 21 October 2003.

62. ITOP 4-2-508, Vehicle Vulnerability Tests Using Mines, 14 April 2005.

63. TOP 2-2-812B, Infrared Signature Measurements of Ground Targets, 4 January 2022.

64. TOP 8-2-111C, Chemical and Biological (CB) Contamination Survivability, Small Items of Equipment, 22 June 2021.

65. ITOP 3-2-836(2.1.1), Main Battle Tank Fire Control Systems – Boresight & Muzzle-Reference – System Alignment/Retention, 31 March 1987.

66. ITOP 3-2-836(2.4.1), Main Battle Tank Fire Control Systems – Computerized Corrections, 29 June 1995.

67. TOP 3-2-813, Field of Firing, 16 December 2009.

68. ITOP 3-2-836(1.3.2.3), Combat Vehicle Fire Control System – Gun/Turret Drive Systems – Frequency Response, 8 April 2002.

69. ITOP 5-2-618, Laser Weapon Danger Areas, 3 June 2010.

70. ITOP 3-2-836(2.2.1), Main Battle Tank Fire Control Systems – Stabilization Accuracy. 29 June 1995.

71. ITOP 3-2-836(2.3.5), Combat Vehicle Fire Control System – Target Tracking, 10 April 2000.

72. ITOP 3-2-836(2.1.2), Main Battle Tank Fire Control Systems – Gun/Sight Synchronization, 31 March 1987.

73. ITOP 3-2-836(1.3.2.2), Combat Vehicle Fire Control System – Gun/Turret Drive Systems – Stabilization, 16 October 2000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

74. ITOP 3-2-836(2.3.4), Combat Vehicle Fire Control System – Target Engagement Times, 26 October 1999.

75. MIL-STD-461E, Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, 20 August 1999.

76. MIL-STD-464D, Electromagnetic Environmental Effects Requirements for Systems, 24 December 2020.

77. TOP 2-2-541, Safe Operation of Mobile Unmanned Ground Vehicle (UGV) Systems, 13 July 2010.

78. TOP 2-2-542, Safe Operation of Weaponized Unmanned Ground Vehicle (UGV) Systems, 013 July 2010.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

APPENDIX D. APPROVAL AUTHORITY.

DEPARTMENT OF THE ARMY
U.S. ARMY WHITE SANDS MISSILE RANGE
100 HEADQUARTERS AVENUE
WHITE SANDS MISSILE RANGE, NEW MEXICO 88002-5000

TEWS-ED

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Test Operations Procedure 02-2-540A Testing of Unmanned Ground Vehicle (UGV) Systems

1. Test Operations Procedure (TOP) 02-2-540A Testing of Unmanned Ground Vehicle (UGV) Systems has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This TOP describes a systematic approach to safety and performance testing of Unmanned Ground Vehicles (UGVs). The objective is to ensure that the design of each UGV includes positive measures to enhance system safety, and that hazards which could reduce system safety are eliminated or controlled to an acceptable level of risk.

3. This document is approved for publication and will be posted to the ASSIST website. The ASSIST website can be accessed at https://assist.dla.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army White Sand Missile Range (TEWS-ED), 100 Headquarters Avenue - Suite 231 White Sands Missile Range, NM 88002-5000; or e-mailed to usarmy.apg.atec.mbx.atec-standards@army.mil.

VINCENT M. LIDDIARD
Executive Director

DISTRIBUTION:
Commander, U.S. Army White Sands Missile Range
Executive Director, U.S. Army Evaluation Center
Commander, U.S. Army Operational Test Command
Commander, U.S. Army Yuma Proving Ground
Commander, U.S. Army Dugway Proving Ground
Commanders, U.S. ATEC Test Centers
Director, U.S. ATEC Tropic Regions Test Center
Director, U.S. ATEC West Desert Test Center

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.

Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to the following address: U.S. Army White Sand Missile Range (TEWS-ED), 100 Headquarters Avenue - Suite 231, White Sands Missile Range, NM 88002-5000.1. Technical information may be obtained from the preparing activity: U.S. Army Aberdeen Test Center (ATC) (TEAT-AD), 6943 Colleran Road, Aberdeen Proving Ground, MD 21005-5059.

Additional copies are available from ASSIST (https://assist.dla.mil/).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:48Z
Check the source to verify that this is the current version before use.